

# Reflection, Symmetry, and Inversion (Homework)

Relations are written below in table form. Reflect each point across the x-axis, and write the coordinates in the tables at the right.

①

|   |    |    |    |    |    |
|---|----|----|----|----|----|
| X | -2 | 11 | 14 | -6 | -1 |
| Y | 9  | -9 | -8 | 4  | 2  |

Reflect across the x-axis →

|   |  |  |  |  |  |
|---|--|--|--|--|--|
| X |  |  |  |  |  |
| Y |  |  |  |  |  |

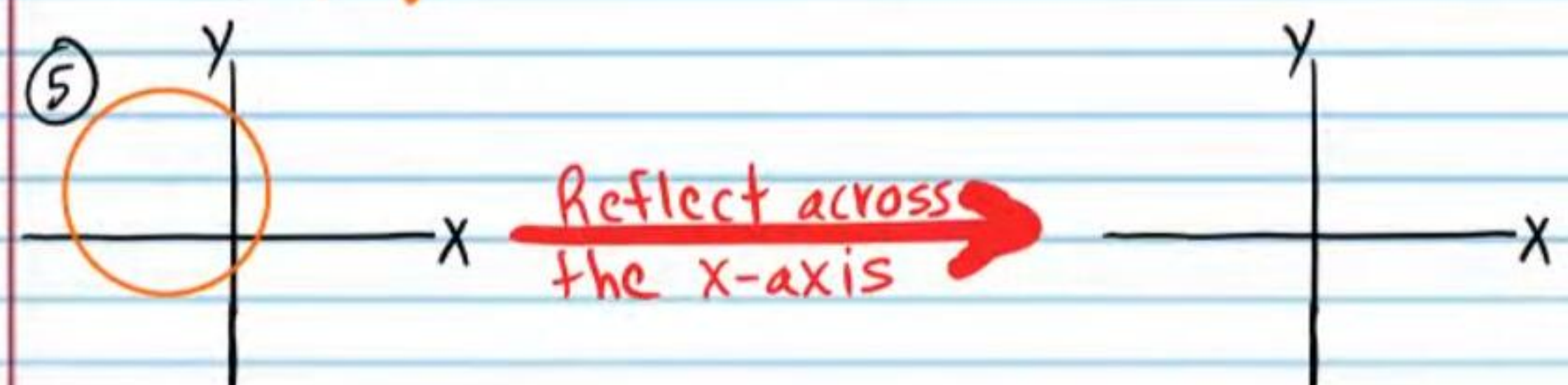
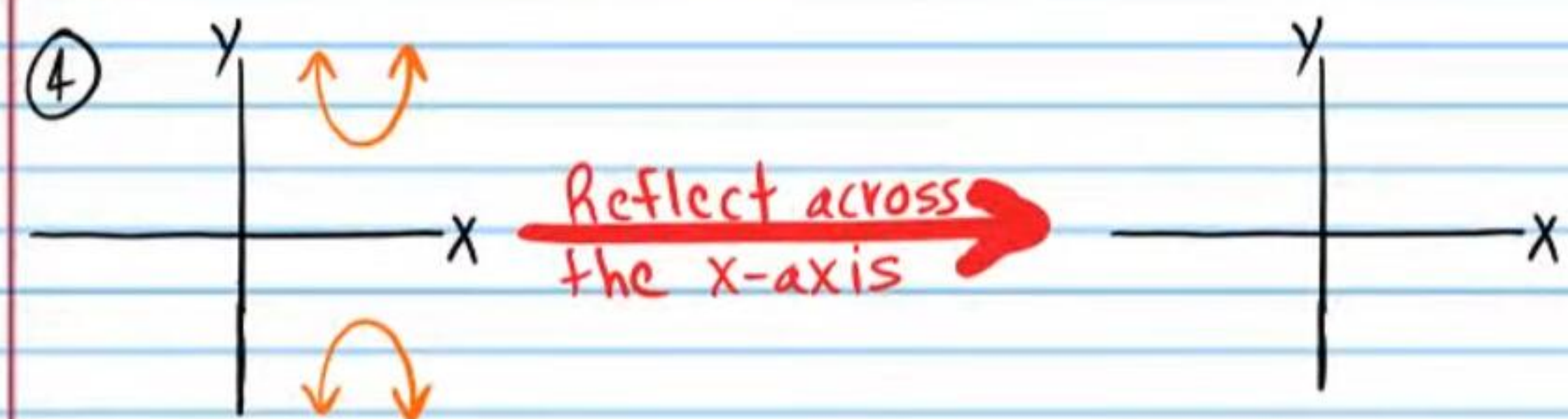
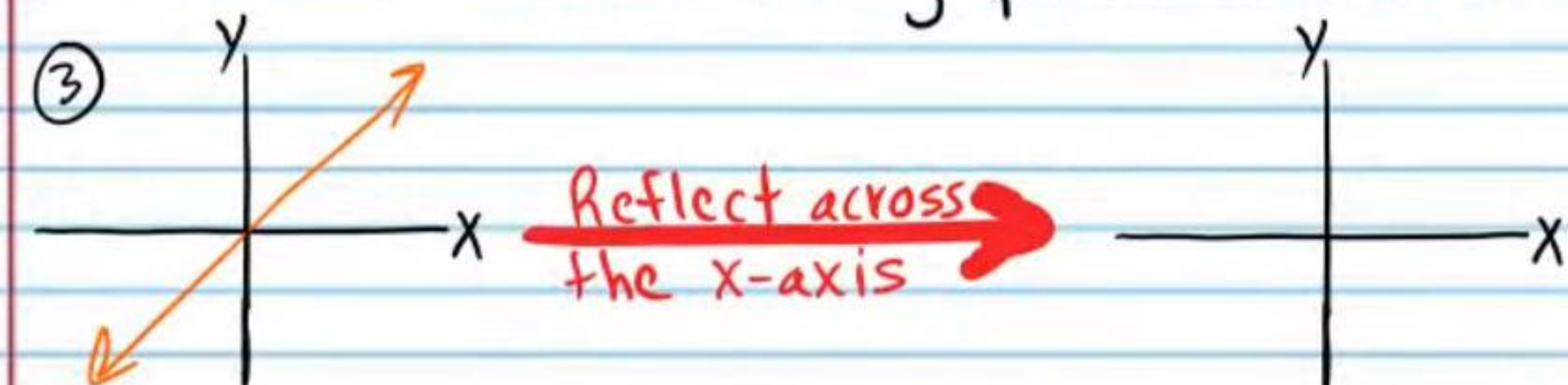
②

|   |    |    |    |    |    |
|---|----|----|----|----|----|
| X | 22 | 21 | 18 | -3 | -3 |
| Y | 0  | 3  | -1 | 6  | 4  |

Reflect across the x-axis →

|   |  |  |  |  |  |
|---|--|--|--|--|--|
| X |  |  |  |  |  |
| Y |  |  |  |  |  |

Relations are written below in the form of graphs. Draw the reflection of each graph across the x-axis.



The equations of relations are written below. Write equations of reflections across the x-axis. Simplify if possible.

⑥  $4x = y$  Reflect across the x-axis →

⑦  $\tan y = x^2$  Reflect across the x-axis →

⑧  $\log_3(2-x) = y$  Reflect across the x-axis →

⑨  $y^4 - |y| + 1 = x$  Reflect across the x-axis →

Relations are written below in table form. Reflect each point across the y-axis, and write the coordinates in the tables at the right.

⑩

|   |    |    |    |    |    |
|---|----|----|----|----|----|
| X | -2 | 11 | 14 | -6 | -1 |
| Y | 9  | -9 | -8 | 4  | 2  |

Reflect across the y-axis →

|   |  |  |  |  |  |
|---|--|--|--|--|--|
| X |  |  |  |  |  |
| Y |  |  |  |  |  |

⑪

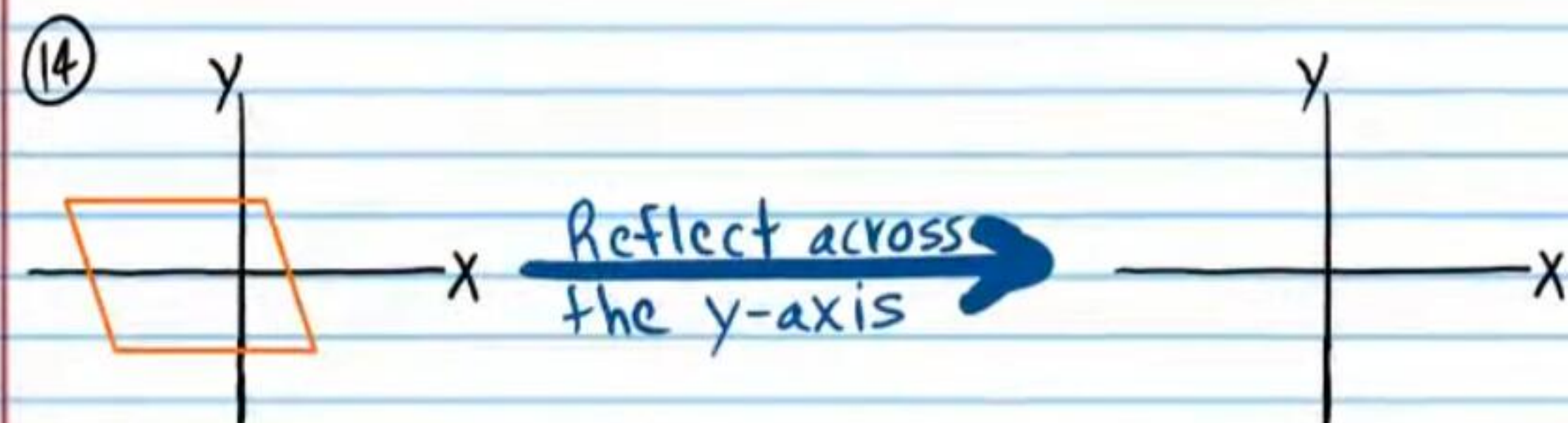
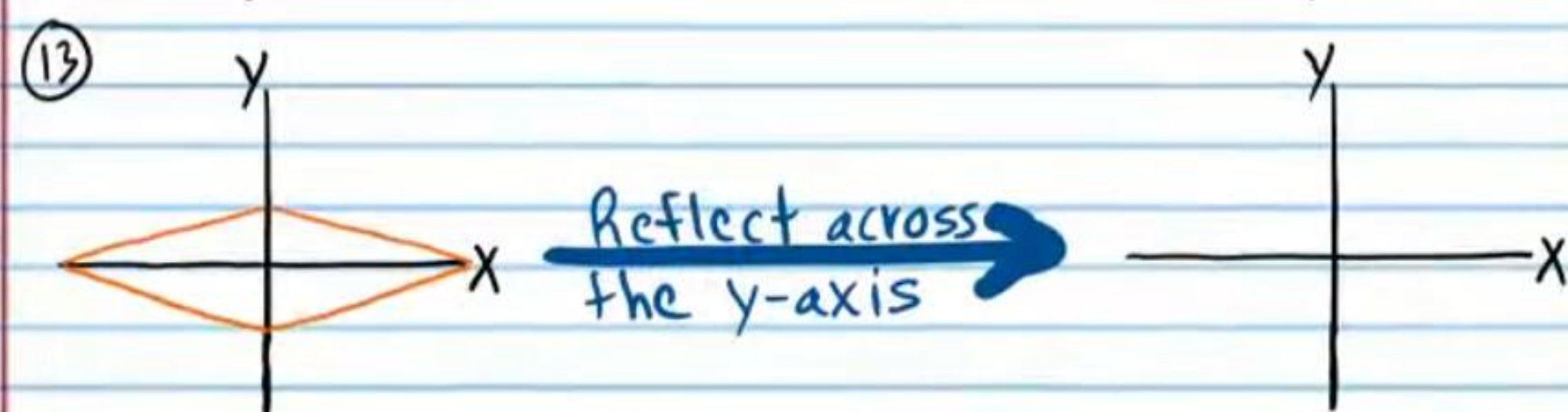
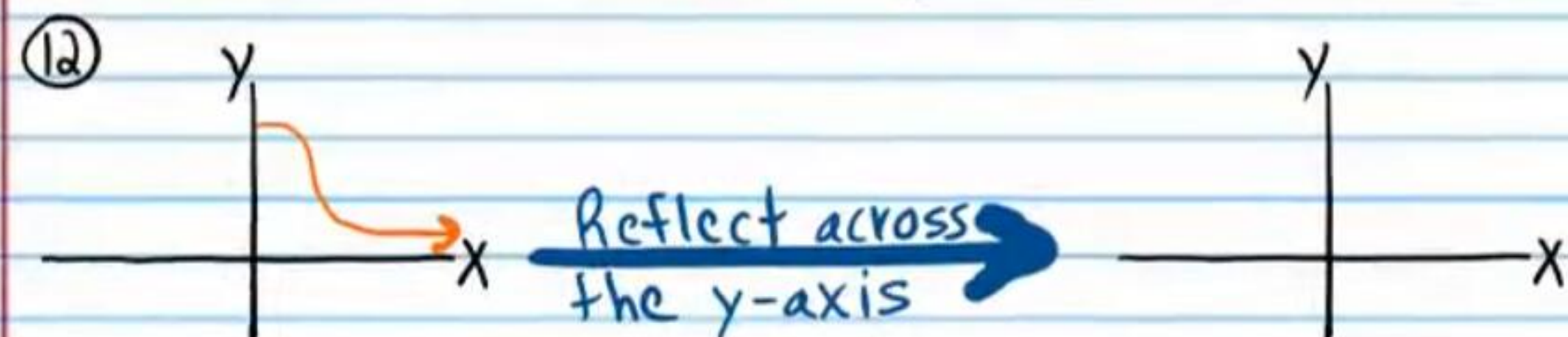
|   |    |    |    |    |    |
|---|----|----|----|----|----|
| X | 22 | 21 | 18 | -3 | -3 |
| Y | 0  | 3  | -1 | 6  | 4  |

Reflect across the y-axis →

|   |  |  |  |  |  |
|---|--|--|--|--|--|
| X |  |  |  |  |  |
| Y |  |  |  |  |  |



Relations are written below in the form of graphs. Draw the reflection of each graph across the y-axis.



The equations of relations are written below. Write equations of reflections across the y-axis. Simplify if possible.

⑮  $10^{16+x} = 2-y$  Reflect across the y-axis →

⑯  $\ln x^2 = \frac{1}{y}$  Reflect across the y-axis →

⑰  $8-2(x+y)=1$  Reflect across the y-axis →

⑱  $10+x^3y^2=3$  Reflect across the y-axis →

Relations are written below in table form. Reflect each point across the origin, and write the coordinates in the tables at the right.

⑲

|   |    |    |    |    |    |
|---|----|----|----|----|----|
| x | -2 | 11 | 14 | -6 | -1 |
| y | 9  | -9 | -8 | 4  | 2  |

Reflect across the origin →

|   |  |  |  |  |  |
|---|--|--|--|--|--|
| x |  |  |  |  |  |
| y |  |  |  |  |  |

⑳

|   |    |    |    |    |    |
|---|----|----|----|----|----|
| x | 22 | 21 | 18 | -3 | -3 |
| y | 0  | 3  | -1 | 6  | 4  |

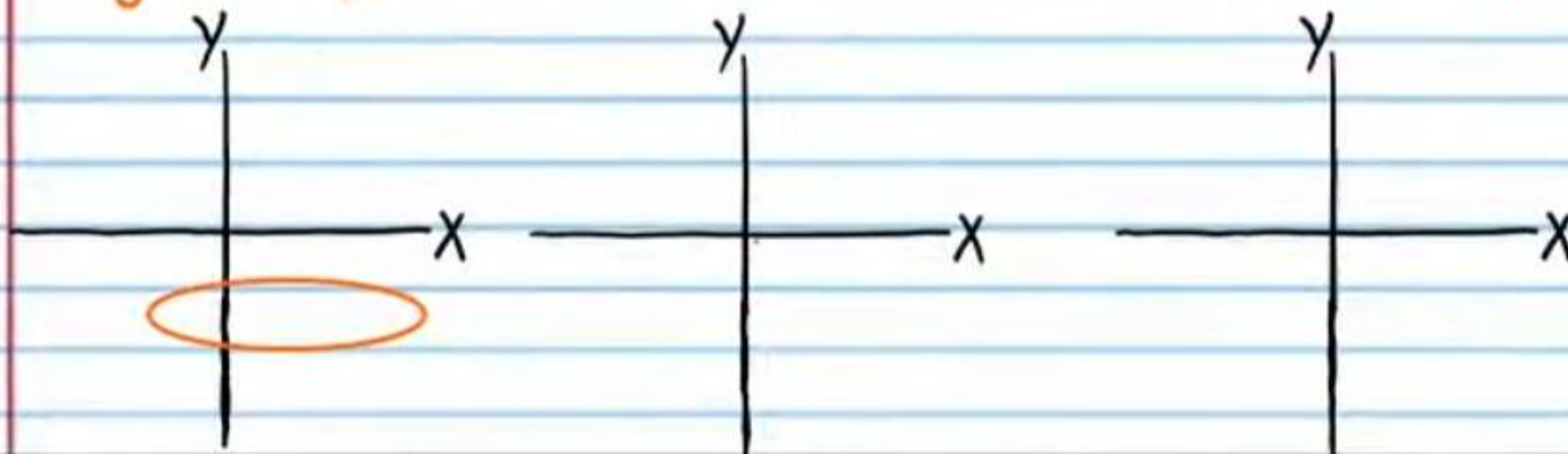
Reflect across the origin →

|   |  |  |  |  |  |
|---|--|--|--|--|--|
| x |  |  |  |  |  |
| y |  |  |  |  |  |

Relations are written below in the form of graphs. Draw the reflection of each graph across the origin.

㉑ Reflect across the origin →

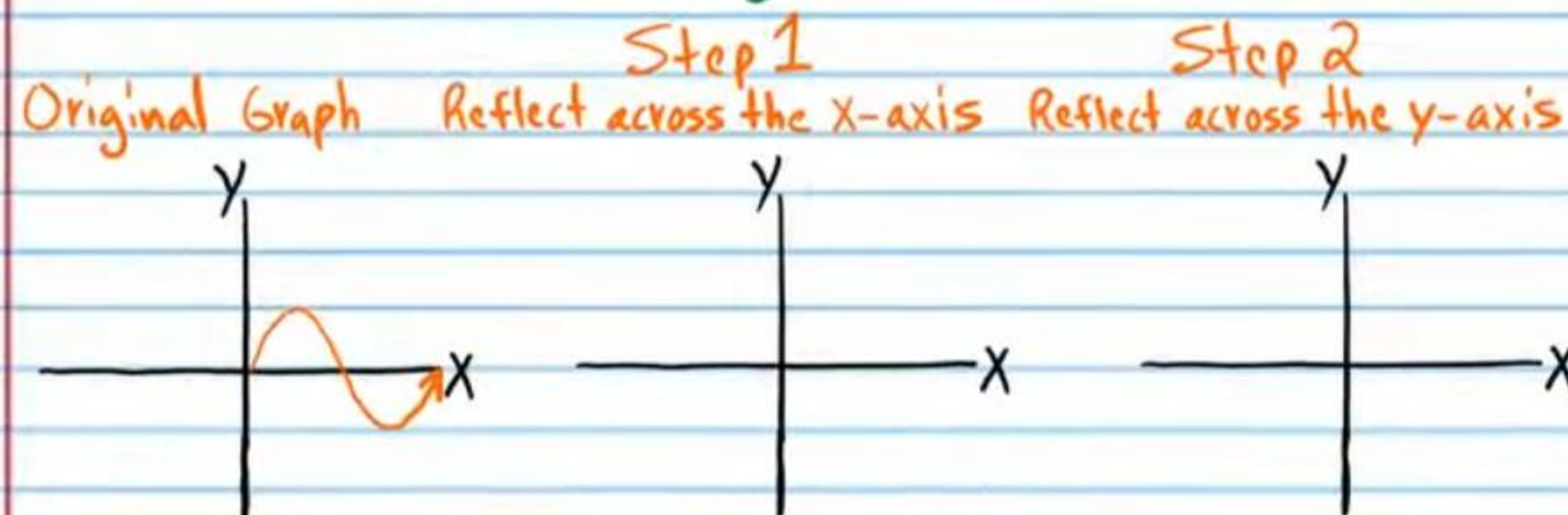
Original Graph    
 Step 1  
Reflect across the x-axis    
 Step 2  
Reflect across the y-axis





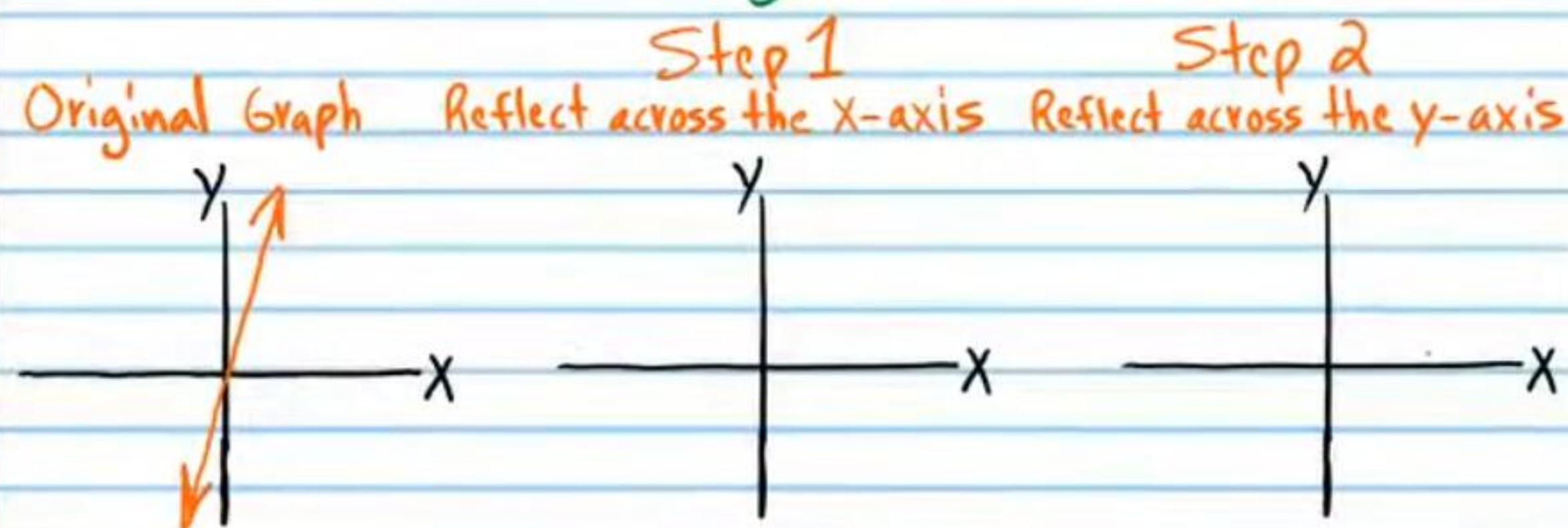
22

Reflect across the origin →



23

Reflect across the origin →



The equations of relations are written below. Write equations of reflections across the origin. Simplify if possible.

24  $x^2 + x + 10 = y$  Reflect across the origin →

25  $y^2 - 7 = \cos(x^2)$  Reflect across the origin →

26  $c^{4x} = |y+x|$  Reflect across the origin →

27  $x = y$  Reflect across the origin →

Each table below has points collected from a graph that is symmetric about the x-axis. Fill the empty spaces with other points from each corresponding graph.

28

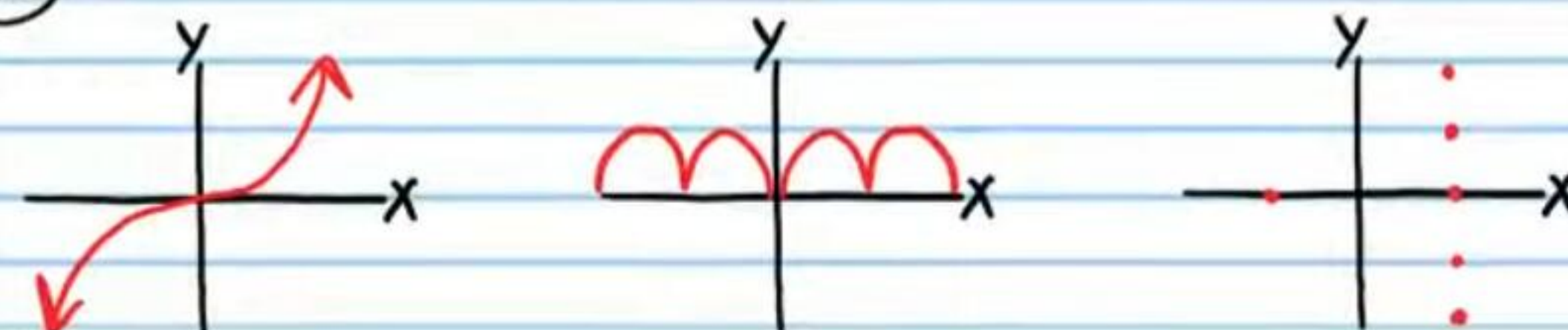
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|---|----|-----|-----|----|-----|--|--|--|--|--|--|--|--|--|
| x | 0  | 87  | -4  | -3 | 5   |  |  |  |  |  |  |  |  |  |
| y | 15 | -20 | 100 | 33 | -14 |  |  |  |  |  |  |  |  |  |

29

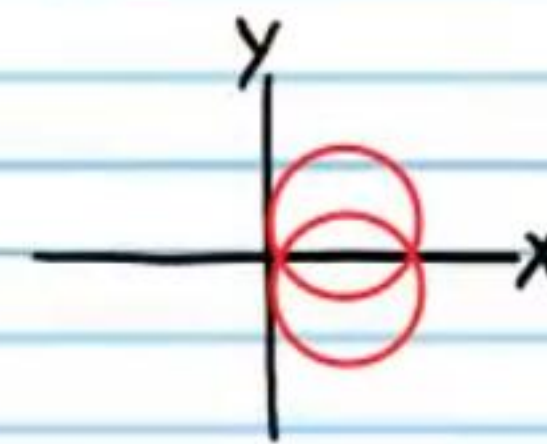
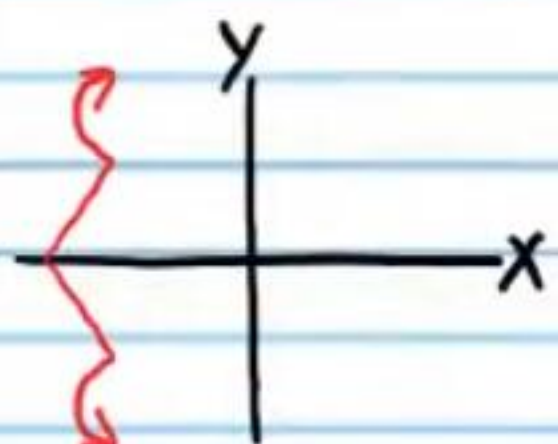
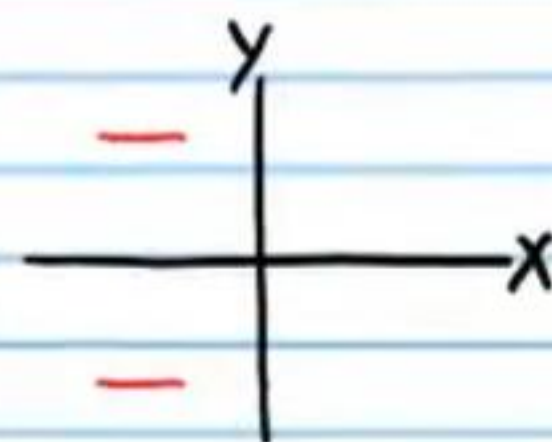
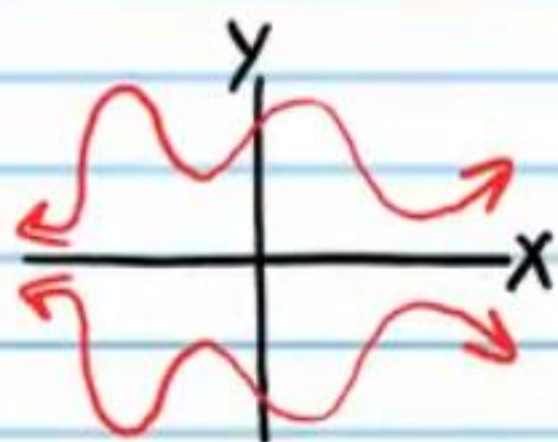
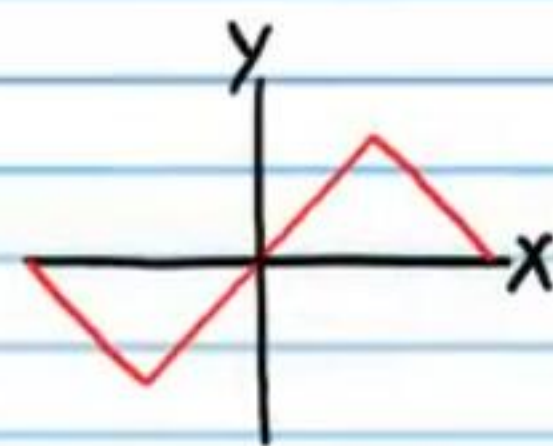
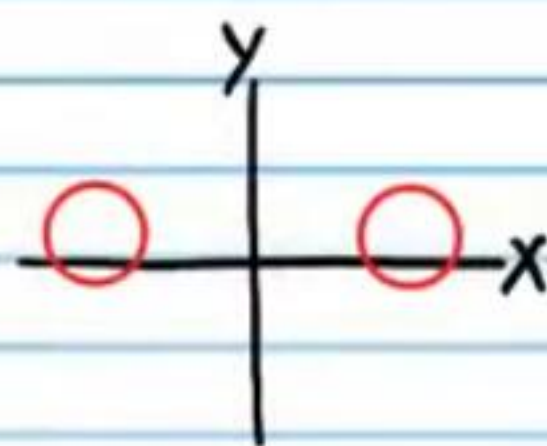
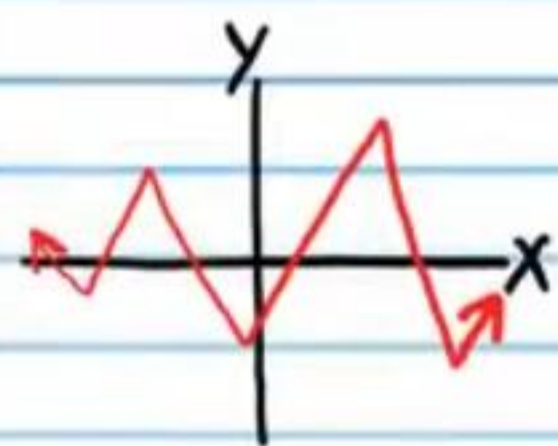
|   |     |    |    |    |     |  |  |  |  |  |  |  |  |  |
|---|-----|----|----|----|-----|--|--|--|--|--|--|--|--|--|
| x | 10  | 34 | 2  | 16 | -19 |  |  |  |  |  |  |  |  |  |
| y | -10 | 30 | -2 | 10 | -10 |  |  |  |  |  |  |  |  |  |

Circle the graphs that appear to be symmetric about the x-axis.

30







Determine if the graph of each relation below is symmetric about the x-axis.

31)  $x + y = 1$

32)  $|x| = |y|$

33)  $\cos x - \sin x = y$

34)  $y - y^3 = 3y^5 + \frac{x}{y}$

Each table below has points collected from a graph that is symmetric about the y-axis. Fill the empty spaces with other points from each corresponding graph.

35)

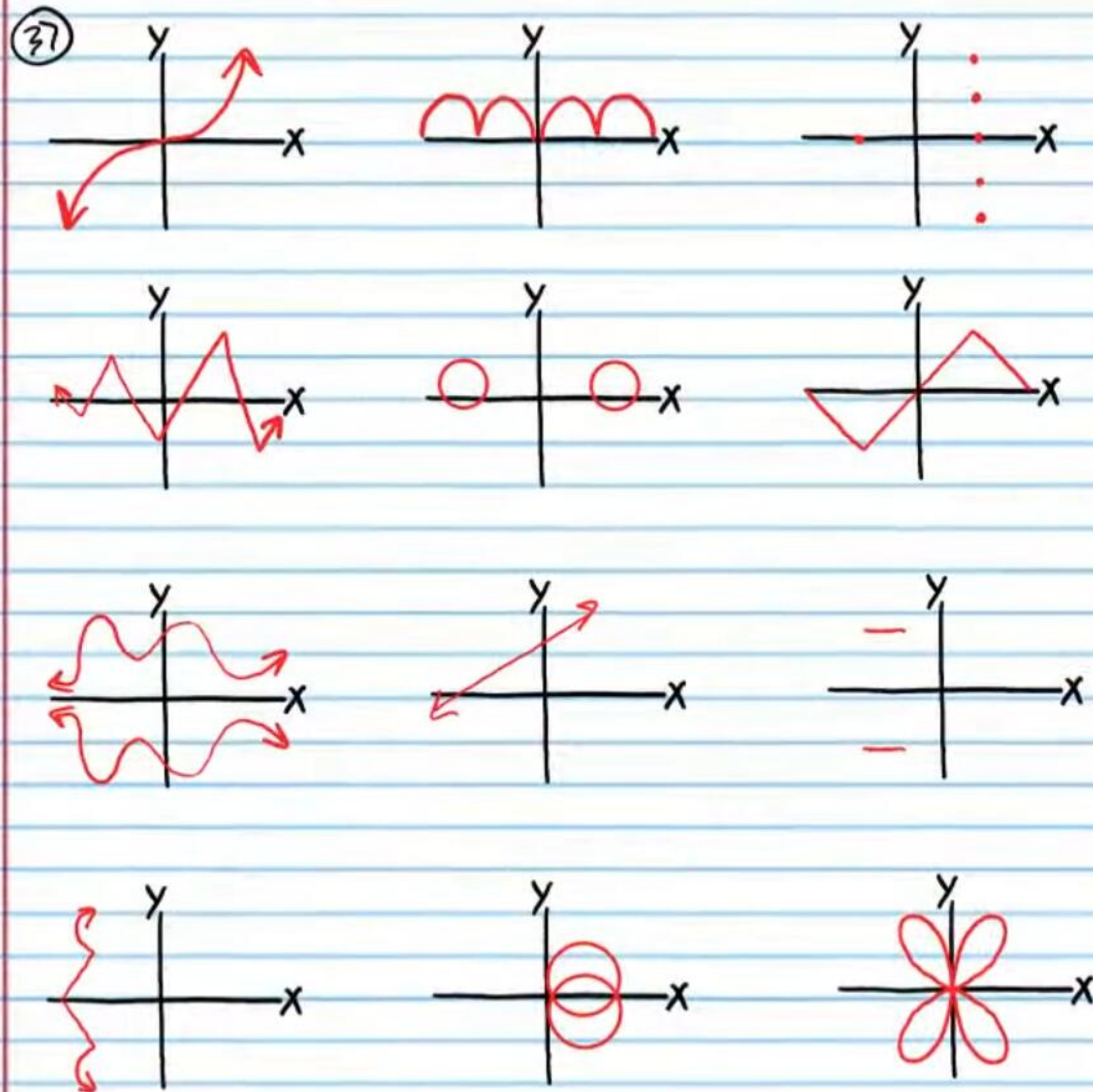
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|---|----|-----|-----|----|-----|--|--|--|--|--|
| x | 0  | 87  | -4  | -3 | 5   |  |  |  |  |  |
| y | 15 | -20 | 100 | 33 | -14 |  |  |  |  |  |

36)

|   |     |    |    |    |     |  |  |  |  |  |
|---|-----|----|----|----|-----|--|--|--|--|--|
| x | 10  | 34 | 2  | 16 | -19 |  |  |  |  |  |
| y | -10 | 30 | -2 | 10 | -10 |  |  |  |  |  |



Circle the graphs that appear to be symmetric about the y-axis.



Determine if the graph of each relation below is symmetric about the y-axis.

38  $\cos x - 2 = y$

39  $2^{x^2} = y$

\* Must use the opposite angle identities.

40  $6x - 3y = |y|$

41  $3x^4 = 1 + y + x$



Each table below has points collected from a graph that is symmetric about the origin. Fill the empty spaces with other points from each corresponding graph.

42

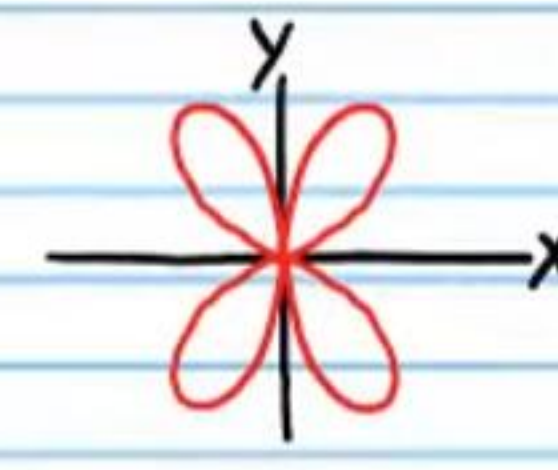
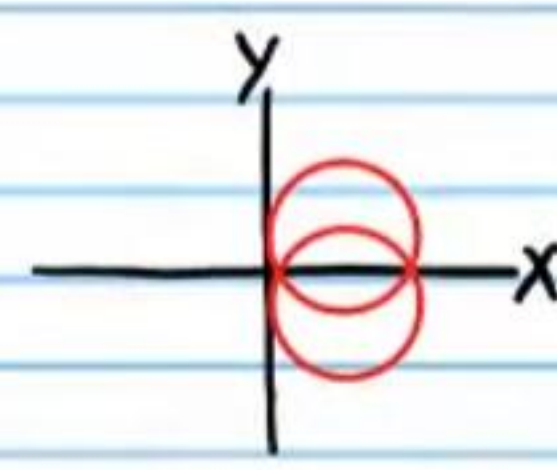
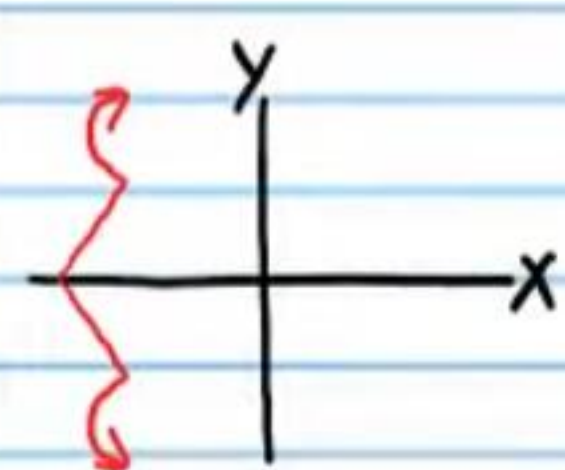
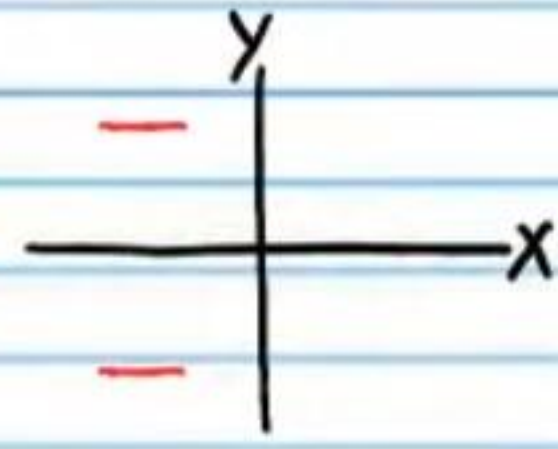
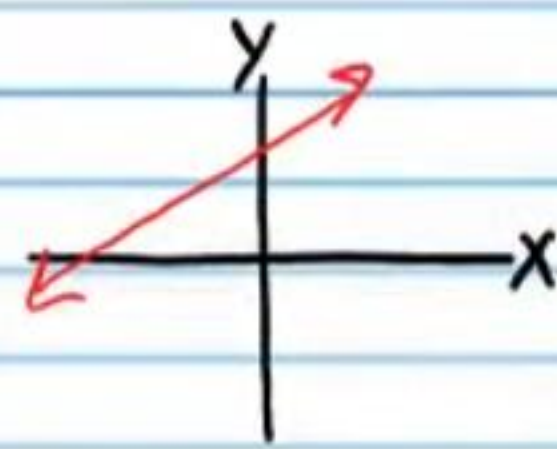
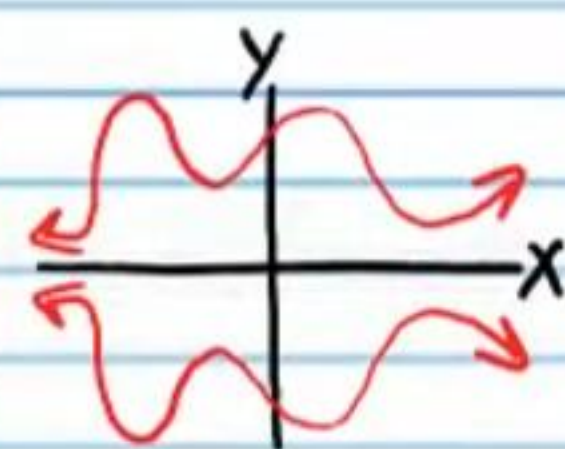
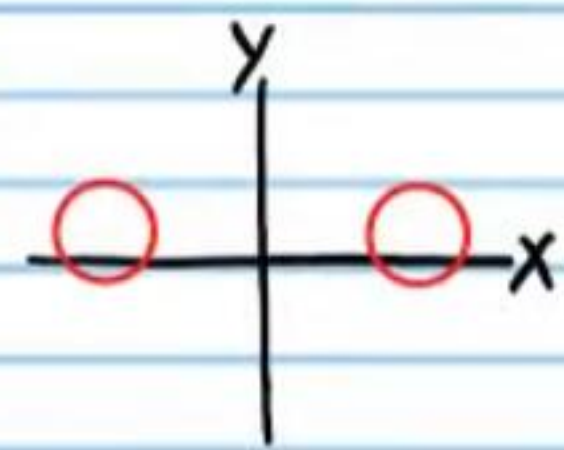
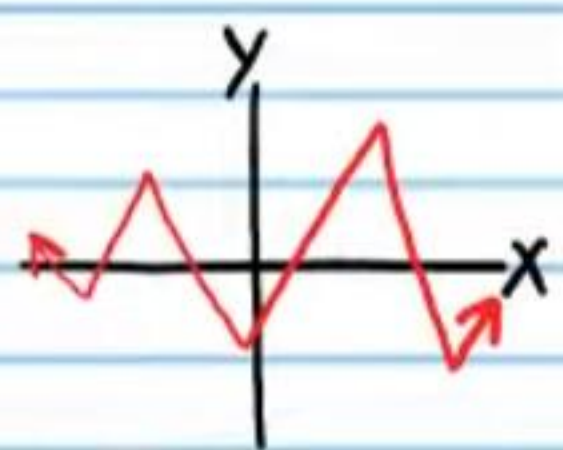
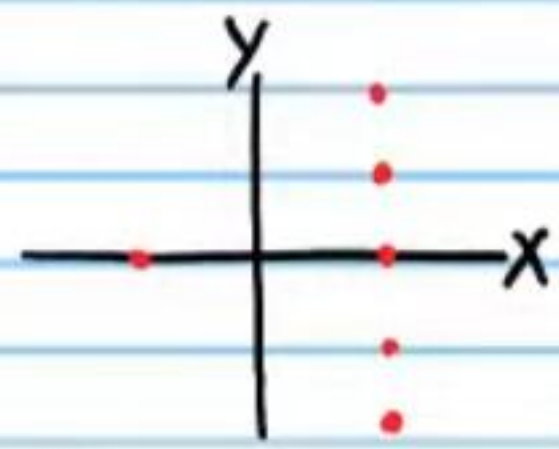
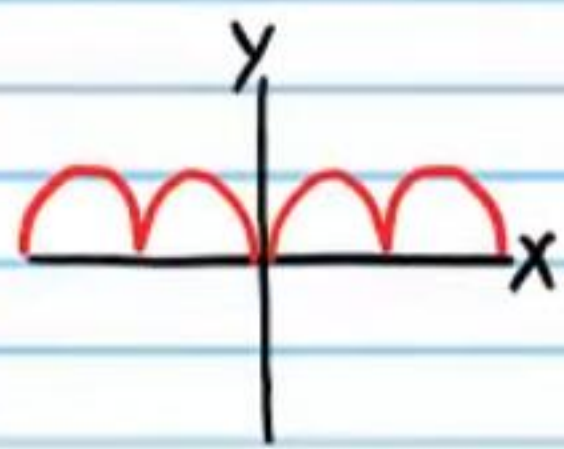
|   |    |     |     |    |     |  |  |  |  |
|---|----|-----|-----|----|-----|--|--|--|--|
| x | 0  | 87  | -4  | -3 | 5   |  |  |  |  |
| y | 15 | -20 | 100 | 33 | -14 |  |  |  |  |

43

|   |     |    |    |    |     |  |  |  |  |
|---|-----|----|----|----|-----|--|--|--|--|
| x | 10  | 34 | 2  | 16 | -19 |  |  |  |  |
| y | -10 | 30 | -2 | 10 | -10 |  |  |  |  |

Circle the graphs that appear to be symmetric about the origin.

44



Determine if the graph of each relation below is symmetric about the origin.

45  $\cos x - \sin x = y$

46  $|x| = |y|$



④⑦  $4\sqrt{x-1} + 2 = y$       ④⑧  $- \csc x + 5y = -x^7$

Relations are written below in table form. Write the coordinates of the corresponding points of the inverse relations (this is reflection across the line  $y=x$ ).

④⑨

|   |    |    |     |    |    |
|---|----|----|-----|----|----|
| X | -5 | 16 | -16 | -6 | -9 |
| Y | -7 | 3  | 2   | 0  | -8 |

Write the inverse relation  
i.e. Reflection across  $y=x$

|   |  |  |  |  |  |
|---|--|--|--|--|--|
| X |  |  |  |  |  |
| Y |  |  |  |  |  |

⑤⑩

|   |    |    |    |    |    |
|---|----|----|----|----|----|
| X | 50 | 60 | 70 | 80 | 90 |
| Y | 40 | 30 | 20 | 10 | 0  |

Write the inverse relation  
i.e. Reflection across  $y=x$

|   |  |  |  |  |  |
|---|--|--|--|--|--|
| X |  |  |  |  |  |
| Y |  |  |  |  |  |

Relations are written below in the form of graphs. Draw the graphs of the inverse relations (i.e. reflect across the line  $y=x$ ).

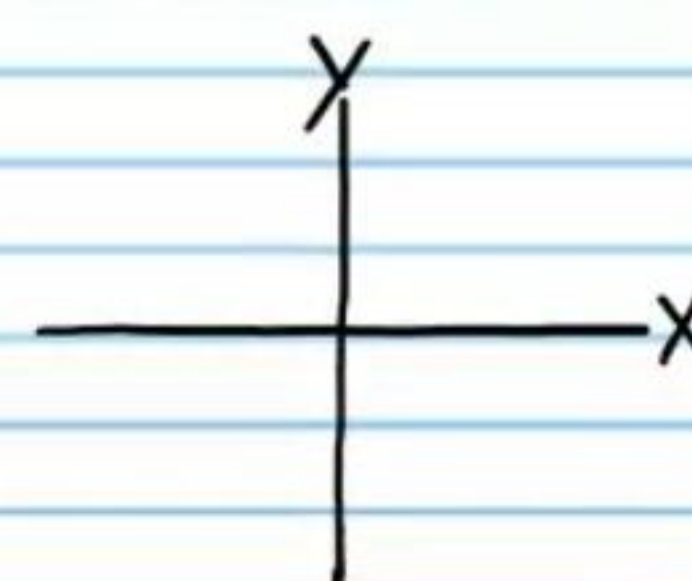
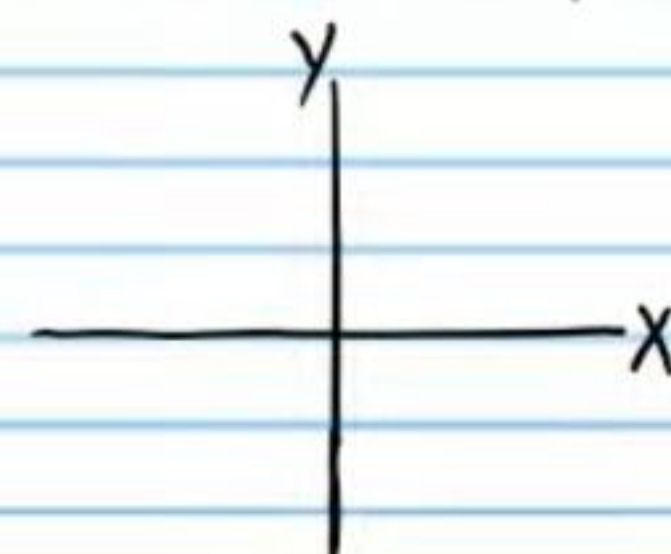
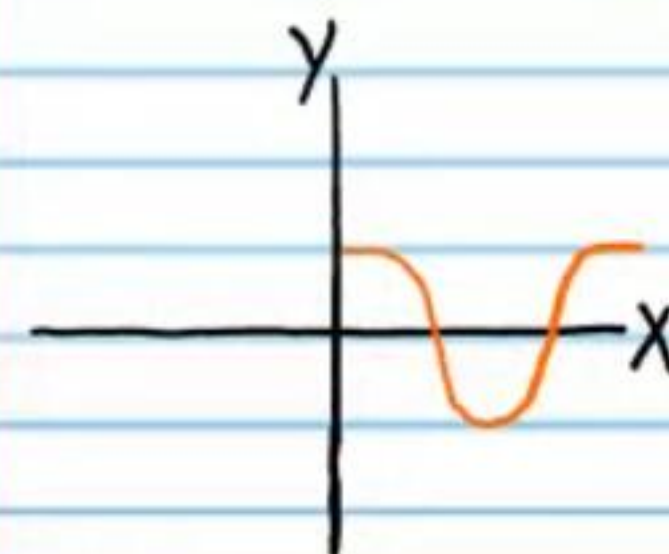
⑤①

Write the inverse relation  
i.e. Reflection across  $y=x$

Step 1

Step 2

Reflect across the  $y$ -axis      Rotate  $90^\circ$  clockwise



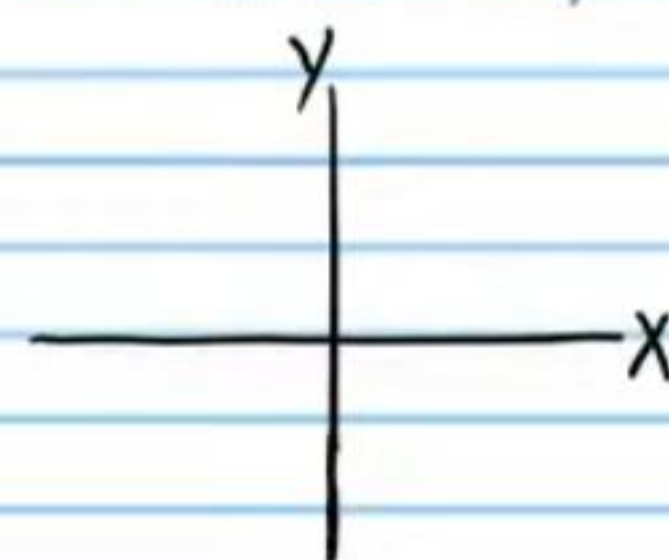
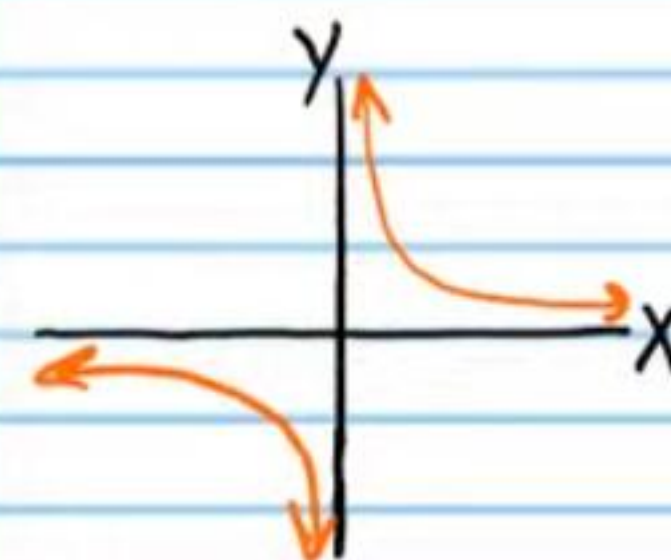
⑤②

Write the inverse relation  
i.e. Reflection across  $y=x$

Step 1

Step 2

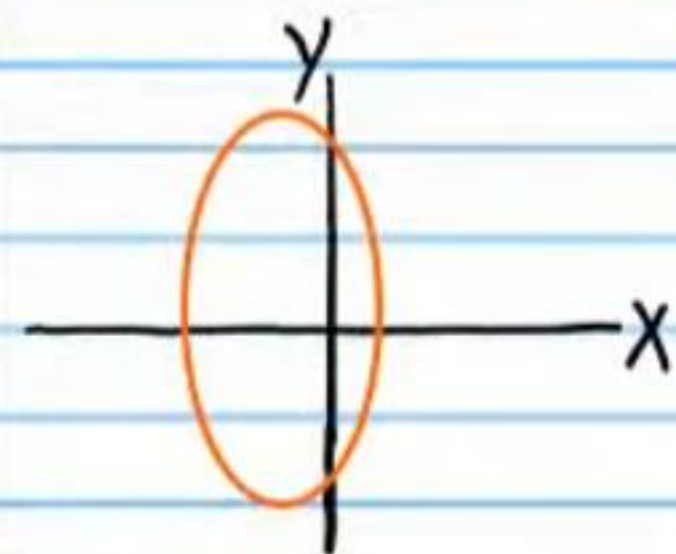
Reflect across the  $y$ -axis      Rotate  $90^\circ$  clockwise



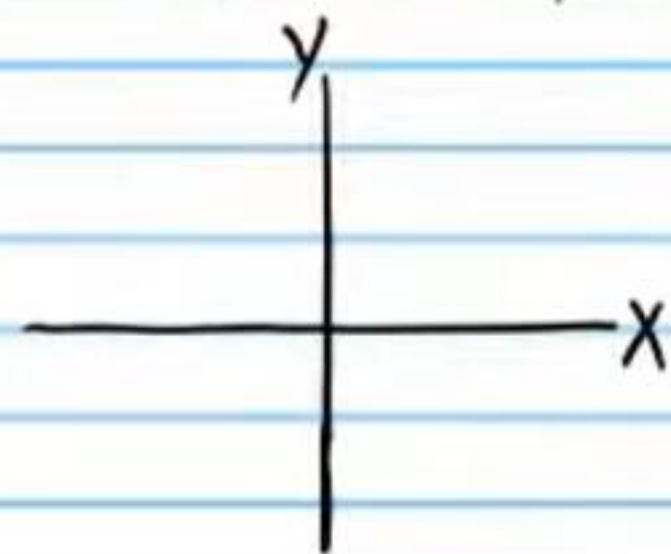


53

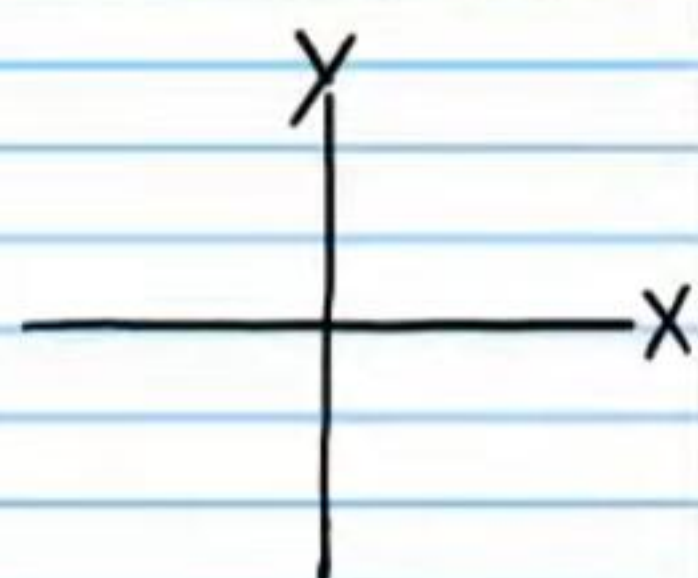
Write the inverse relation  
i.e. Reflection across  $y=x$  →



Step 1  
Reflect across the y-axis



Step 2  
Rotate 90° clockwise



The equations of relations are written below.  
Write equations of the inverse relations.

54  $x + y = 1$

Write the inverse relation  
i.e. Reflection across  $y=x$  →

55  $|x| = |y|$

Write the inverse relation  
i.e. Reflection across  $y=x$  →

56  $x^3 + 10x^2 = y^2$

Write the inverse relation  
i.e. Reflection across  $y=x$  →

57  $4^y - 2^{y-1} = x$

Write the inverse relation  
i.e. Reflection across  $y=x$  →



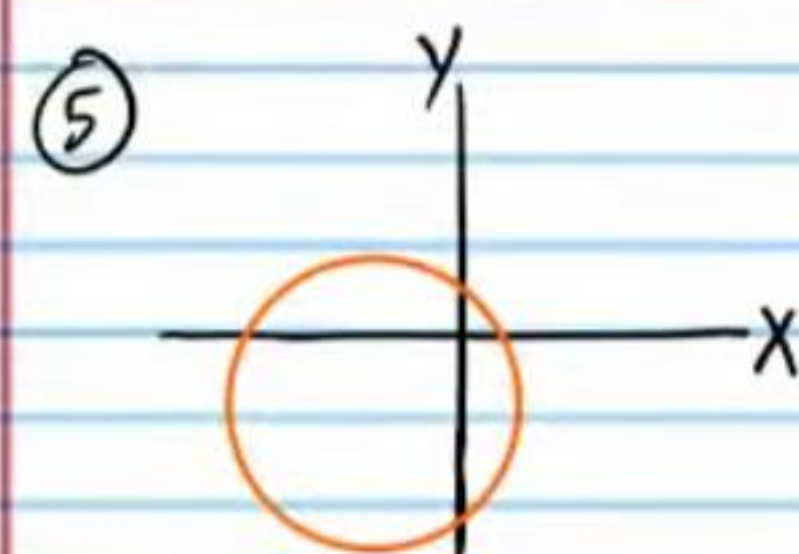
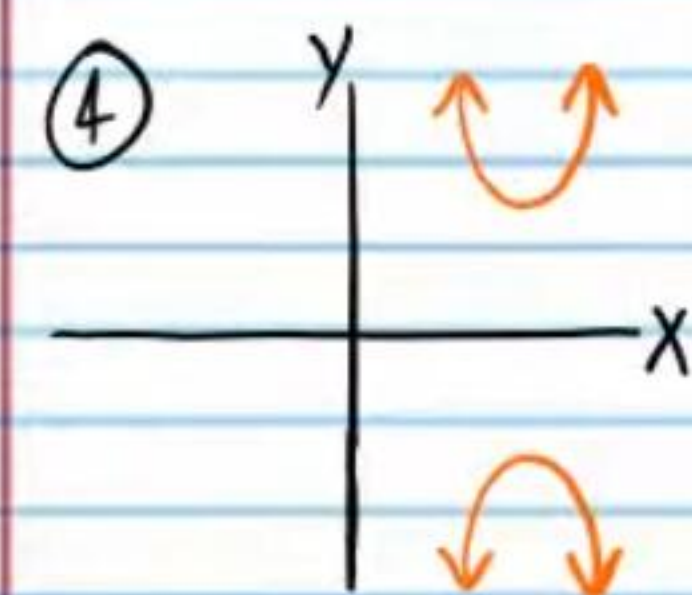
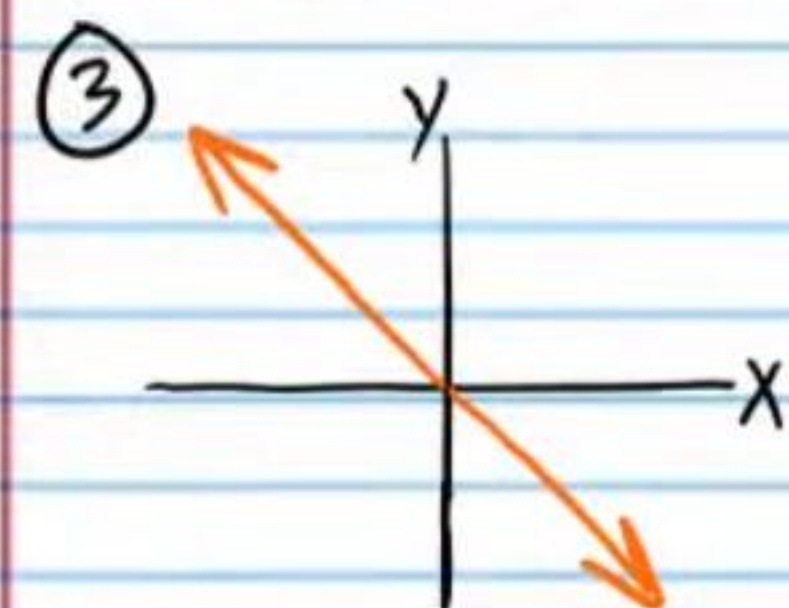
# Answers

①

|   |    |    |    |    |    |
|---|----|----|----|----|----|
| x | -2 | 11 | 14 | -6 | -1 |
| y | -9 | 9  | 8  | -4 | -2 |

②

|   |    |    |    |    |    |
|---|----|----|----|----|----|
| x | 22 | 21 | 18 | -3 | -3 |
| y | 0  | -3 | 1  | -6 | -4 |



⑥  $4x = -y$

⑦  $\tan(-y) = x^2$  or  $-\tan y = x^2$

⑧  $\log_3(2-x) = -y$

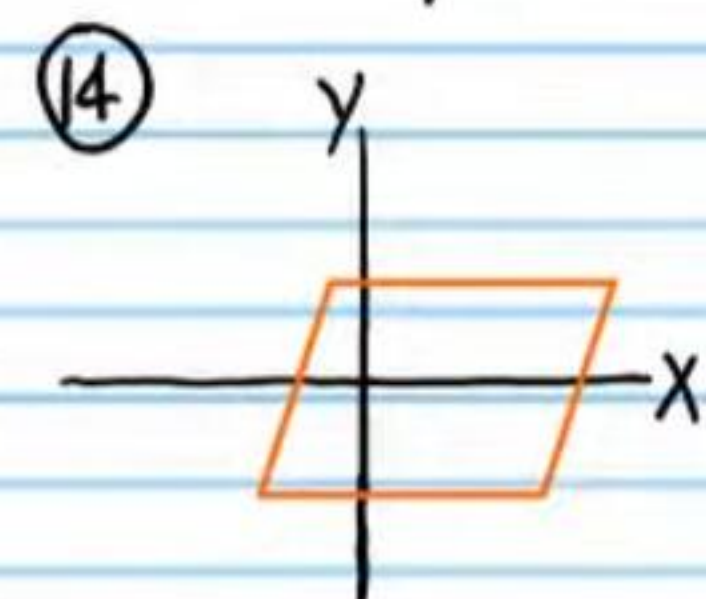
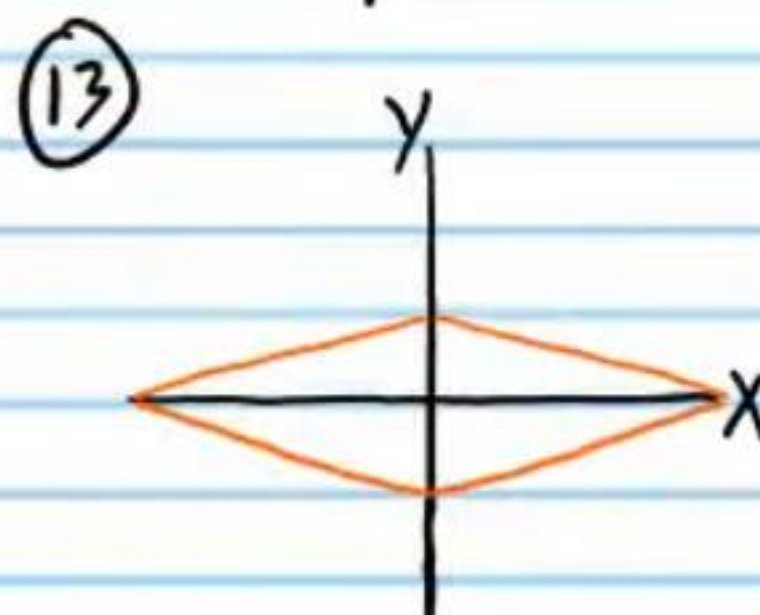
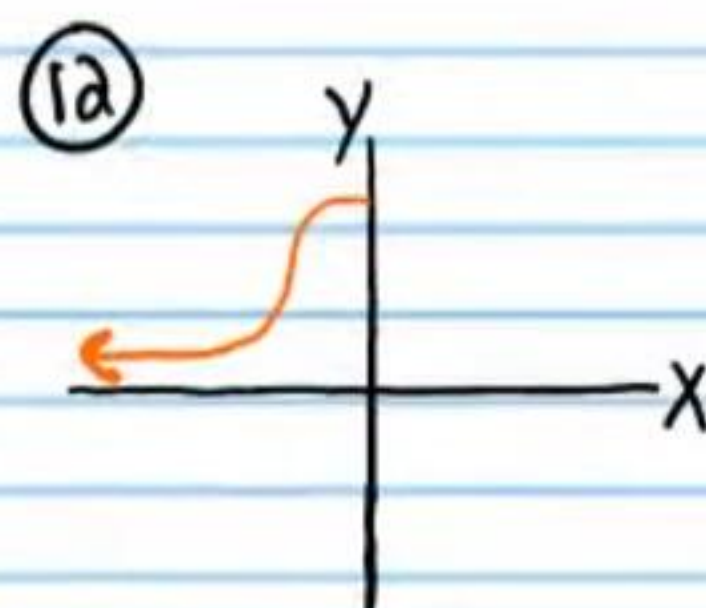
⑨  $y^4 - |y| + 1 = x$

⑩

|   |   |     |     |   |   |
|---|---|-----|-----|---|---|
| x | 2 | -11 | -14 | 6 | 1 |
| y | 9 | -9  | -8  | 4 | 2 |

⑪

|   |     |     |     |   |   |
|---|-----|-----|-----|---|---|
| x | -22 | -21 | -18 | 3 | 3 |
| y | 0   | 3   | -1  | 6 | 4 |



⑮  $10^{16-x} = 2-y$

⑯  $\ln x^2 = \frac{1}{y}$

⑰  $8 - 2(-x+y) = 1$

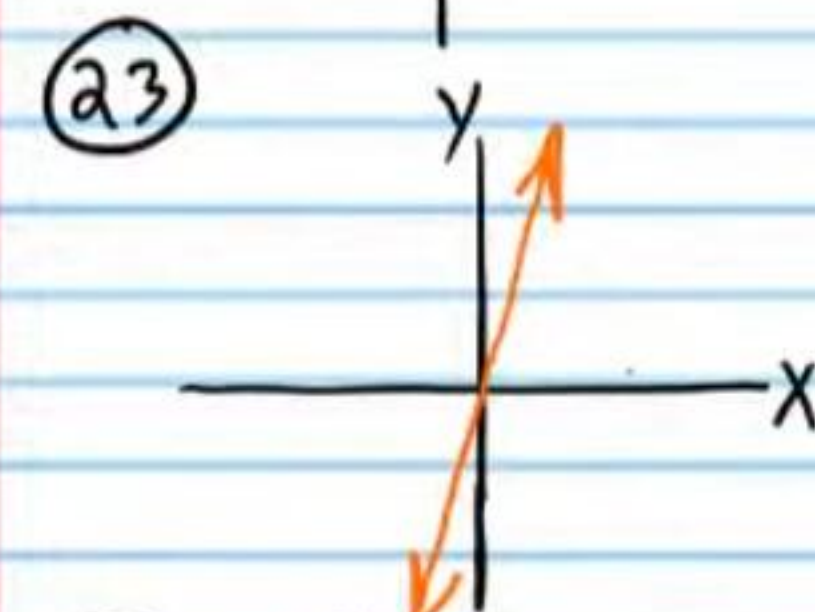
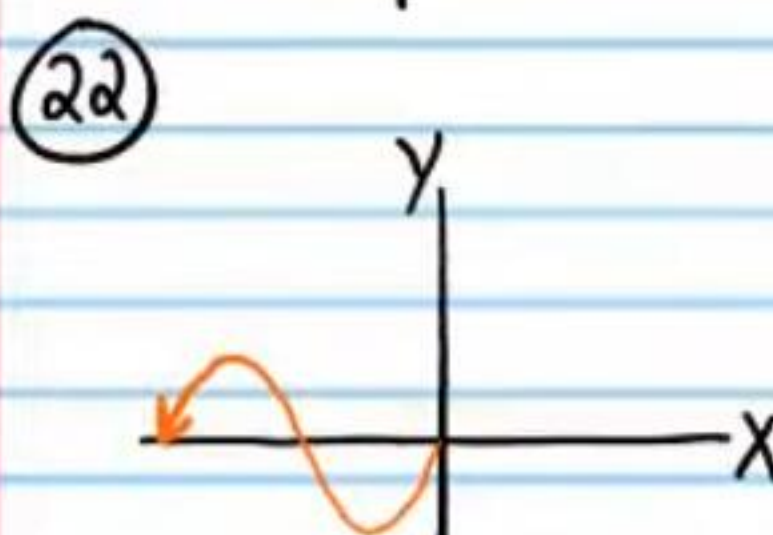
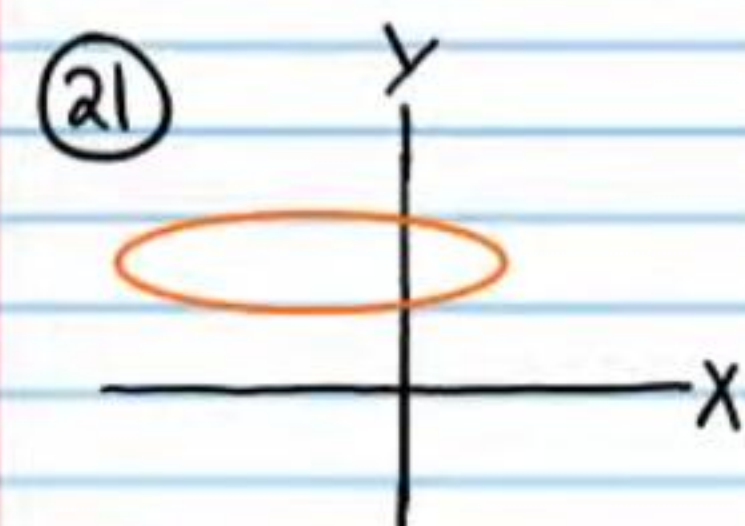
⑱  $10 - x^3 y^2 = 3$

⑲

|   |    |     |     |    |    |
|---|----|-----|-----|----|----|
| x | 2  | -11 | -14 | 6  | 1  |
| y | -9 | 9   | 8   | -4 | -2 |

⑳

|   |     |     |     |    |    |
|---|-----|-----|-----|----|----|
| x | -22 | -21 | -18 | 3  | 3  |
| y | 0   | -3  | 1   | -6 | -4 |



㉔  $x^2 - x + 10 = -y$

㉕  $y^2 - 7 = \cos(x^2)$



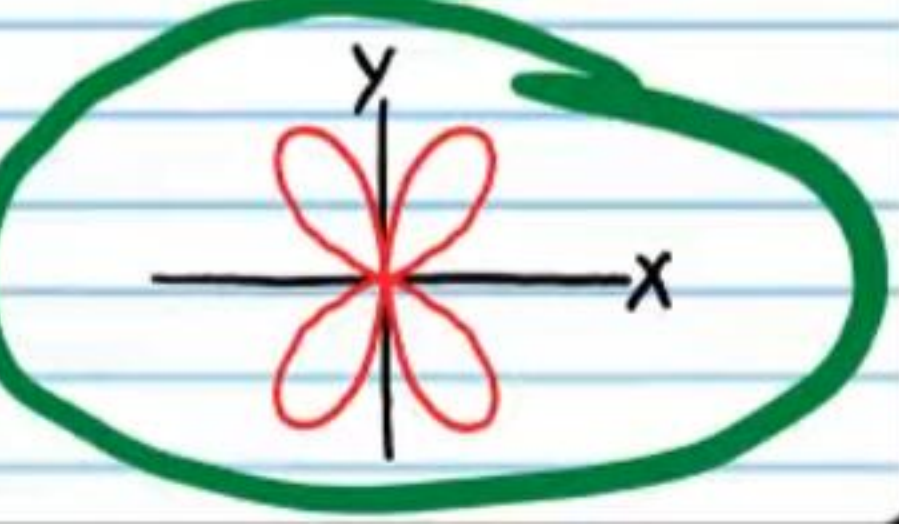
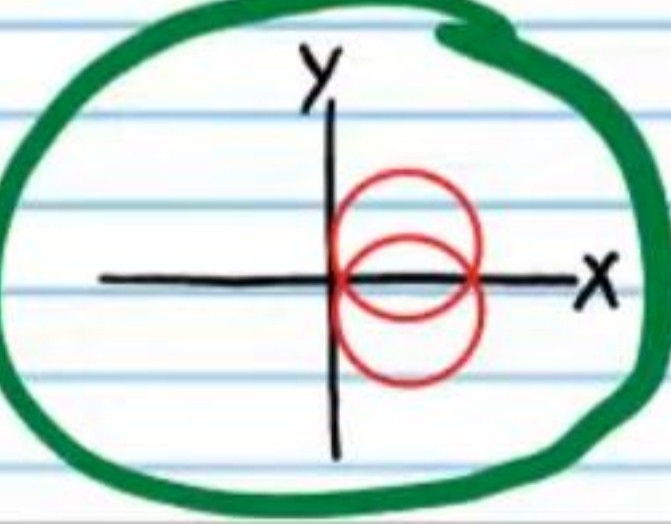
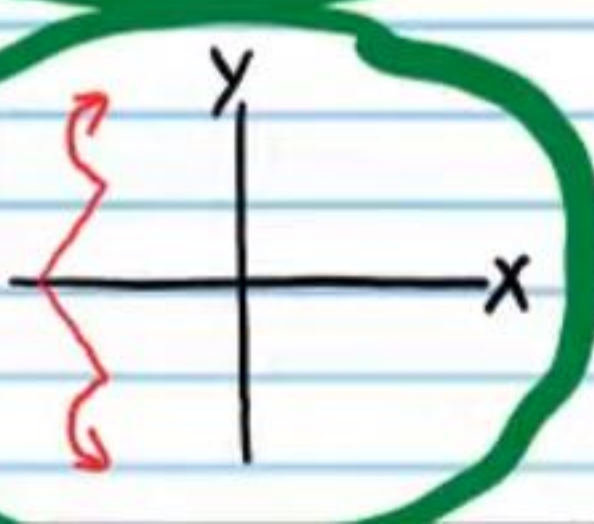
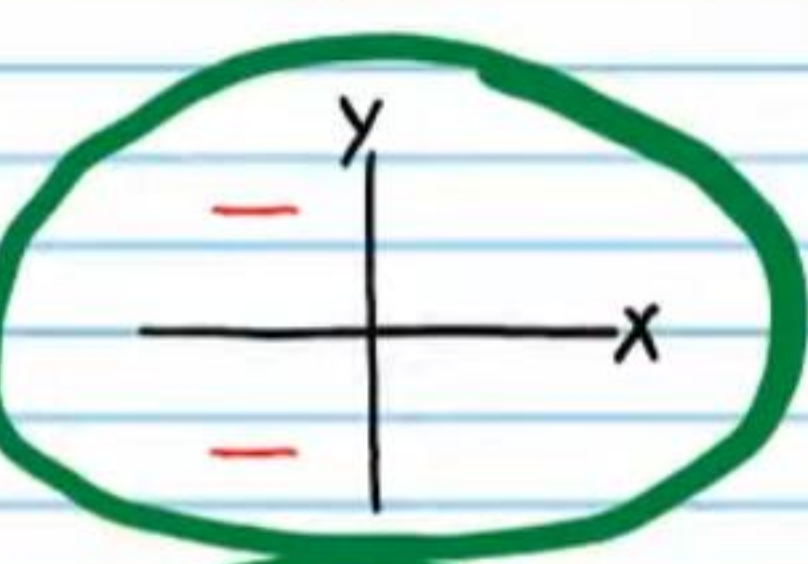
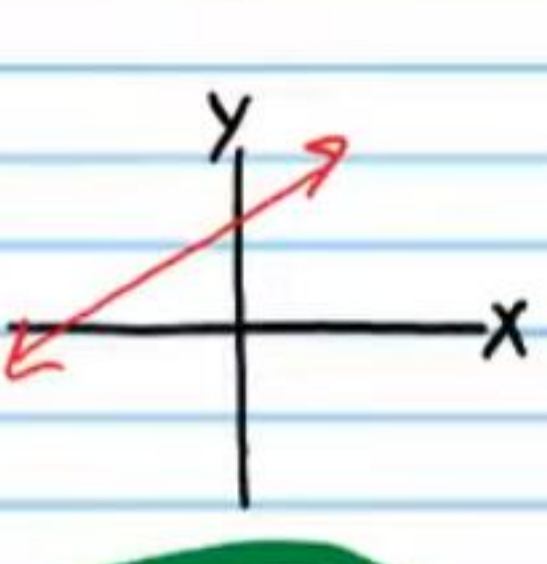
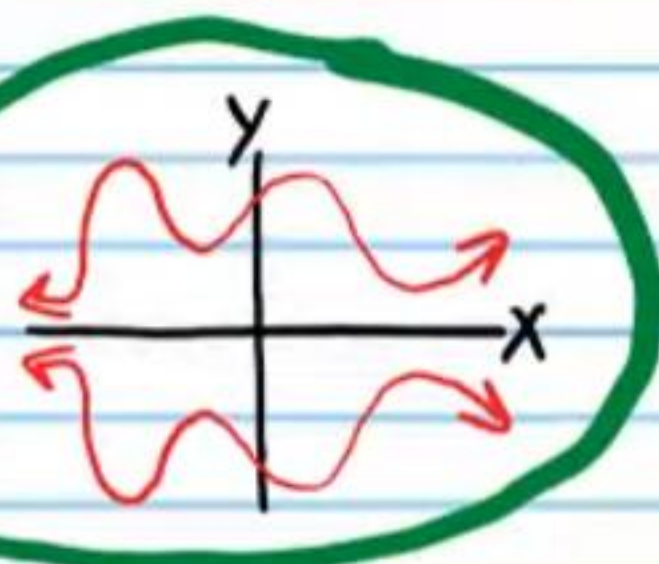
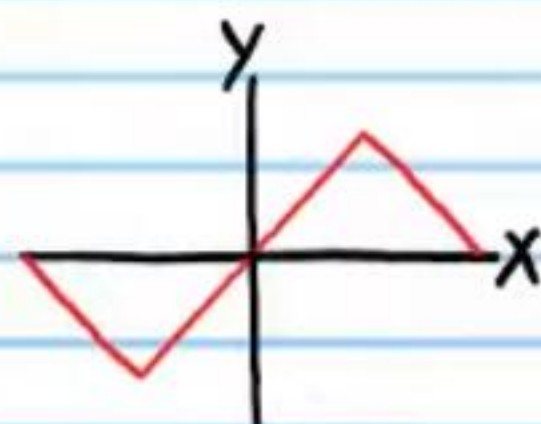
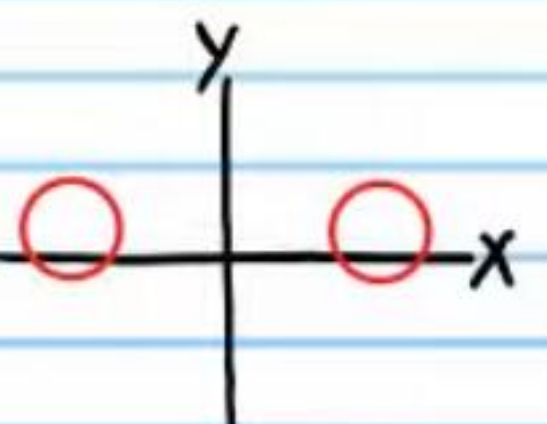
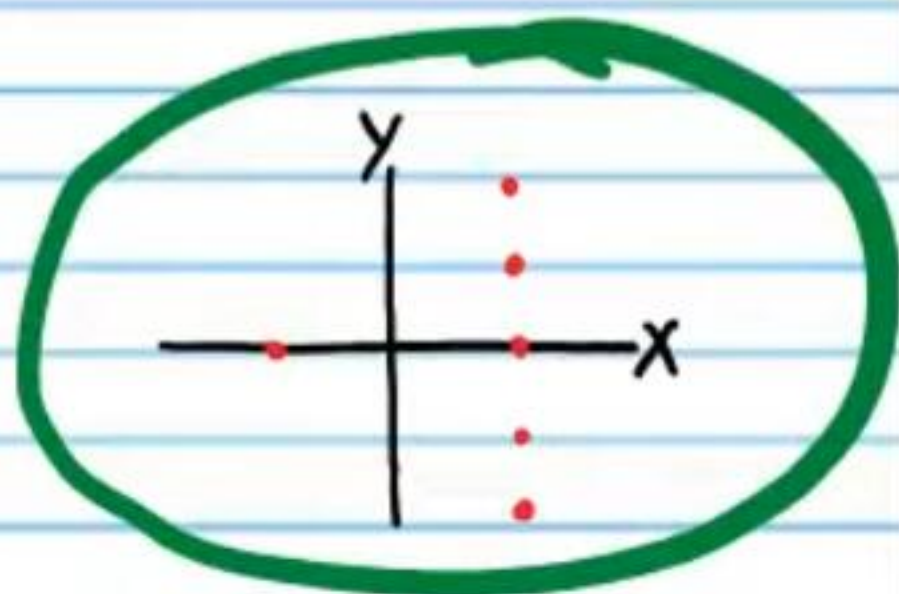
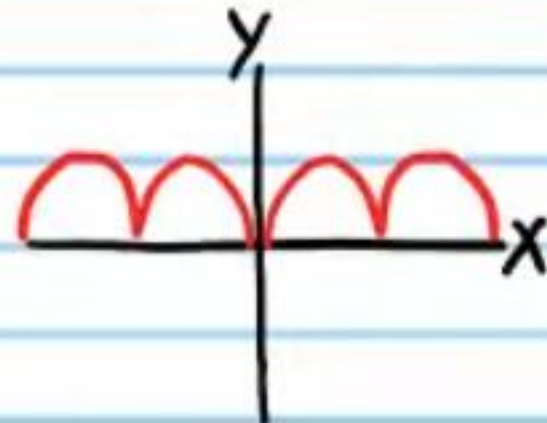
$$(26) e^{-4x} = |-y-x| \text{ or } e^{-4x} = |y+x|$$

$$(27) x=y$$

$$(28) \begin{array}{|c|c|c|c|c|c|c|c|c|c|c|c|} \hline x & 0 & 87 & -4 & -3 & 5 & 0 & 87 & -4 & -3 & 5 \\ \hline y & 15 & -20 & 100 & 33 & -14 & -15 & 20 & -100 & -33 & 14 \\ \hline \end{array}$$

$$(29) \begin{array}{|c|c|c|c|c|c|c|c|c|c|c|c|} \hline x & 10 & 34 & 2 & 16 & -19 & 10 & 34 & 2 & 16 & -19 \\ \hline y & -10 & 30 & -2 & 10 & -10 & 10 & -30 & 2 & -10 & 10 \\ \hline \end{array}$$

(30)



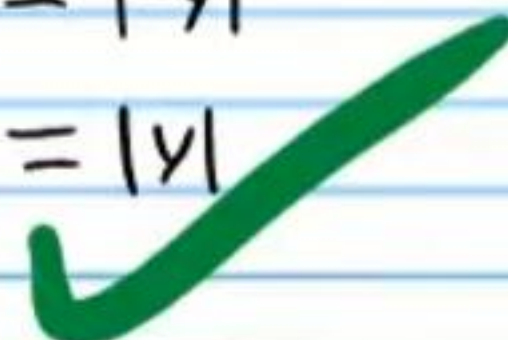
$$(31) x-y=1$$



$$(32) |x|=|y|$$

$$|x|=|-y|$$

$$|x|=|y|$$



$$(33) \cos x - \sin x = y$$

$$\cos x - \sin x = -y$$



$$(34) y - y^3 = 3y^5 + \frac{x}{y}$$

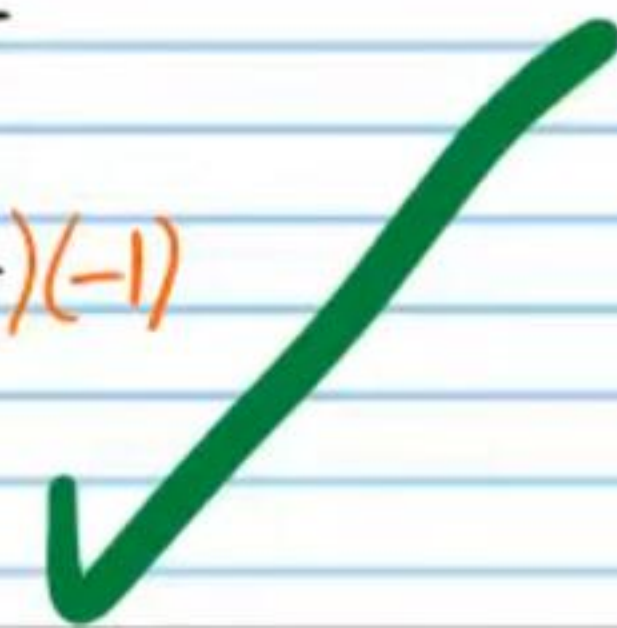
$$-y - (-y)^3 = 3(-y)^5 + \frac{x}{-y}$$

$$-y - (-y^3) = 3(-y^5) - \frac{x}{y}$$

$$-y + y^3 = -3y^5 - \frac{x}{y}$$

$$-1(-y + y^3) = (-3y^5 - \frac{x}{y})(-1)$$

$$y - y^3 = 3y^5 + \frac{x}{y}$$



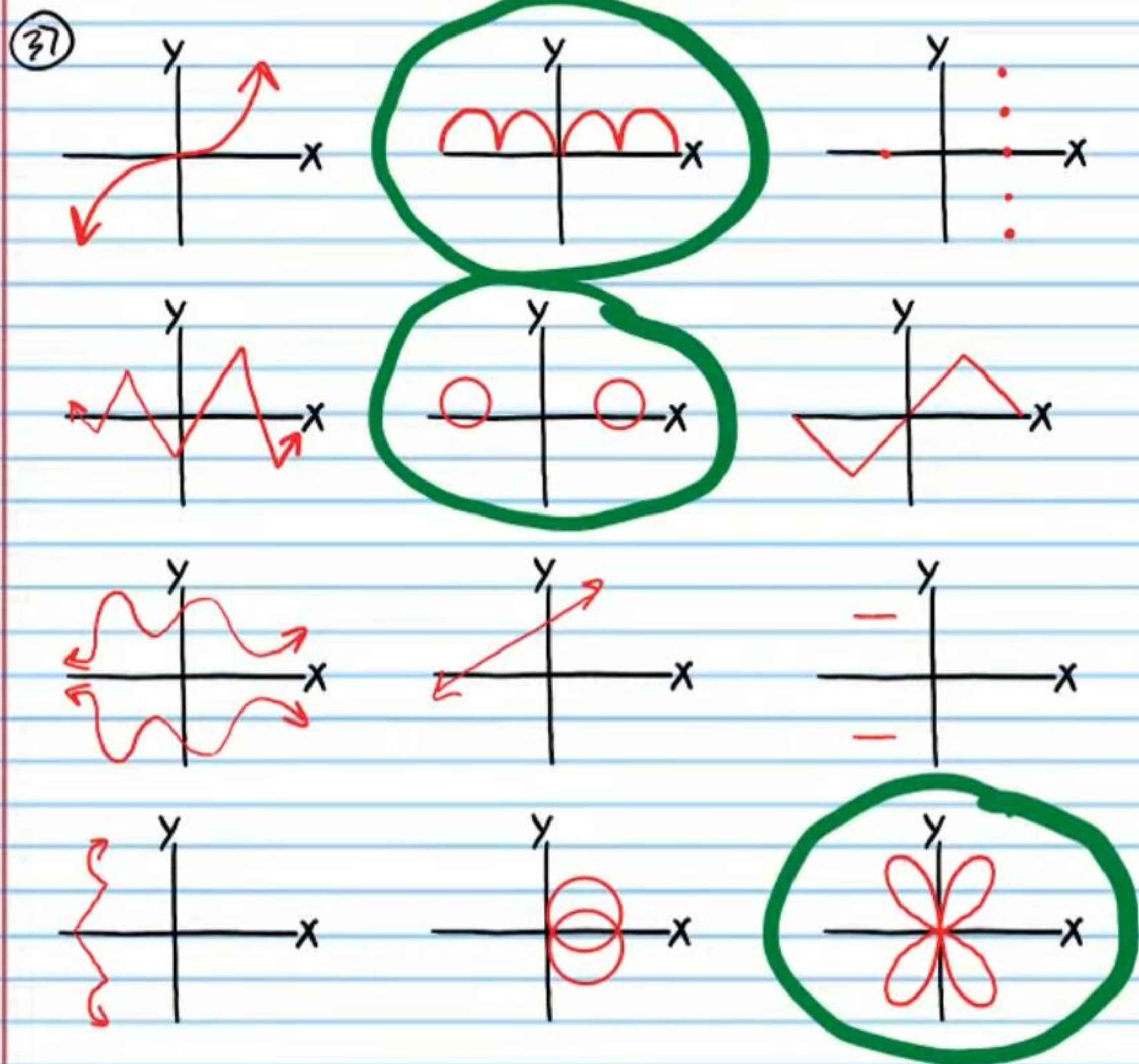


35

|   |    |     |     |    |     |    |     |     |    |     |
|---|----|-----|-----|----|-----|----|-----|-----|----|-----|
| X | 0  | 87  | -4  | -3 | 5   | 0  | -87 | 4   | 3  | -5  |
| Y | 15 | -20 | 100 | 33 | -14 | 15 | -20 | 100 | 33 | -14 |

36

|   |     |    |    |    |     |     |     |    |     |     |
|---|-----|----|----|----|-----|-----|-----|----|-----|-----|
| X | 10  | 34 | 2  | 16 | -19 | -10 | -34 | -2 | -16 | 19  |
| Y | -10 | 30 | -2 | 10 | -10 | -10 | 30  | -2 | 10  | -10 |



38  $\cos x - 2 = y$



39  $2^{x^2} = y$



40  $-6x - 3y = |y|$



41  $3x^4 = 1 + y - x$

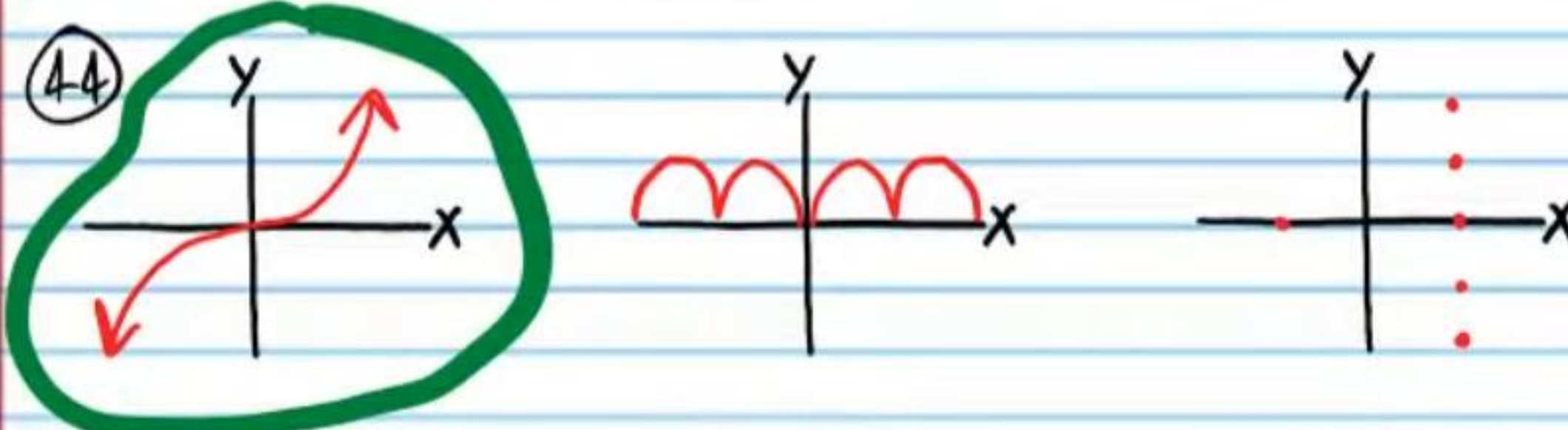


42

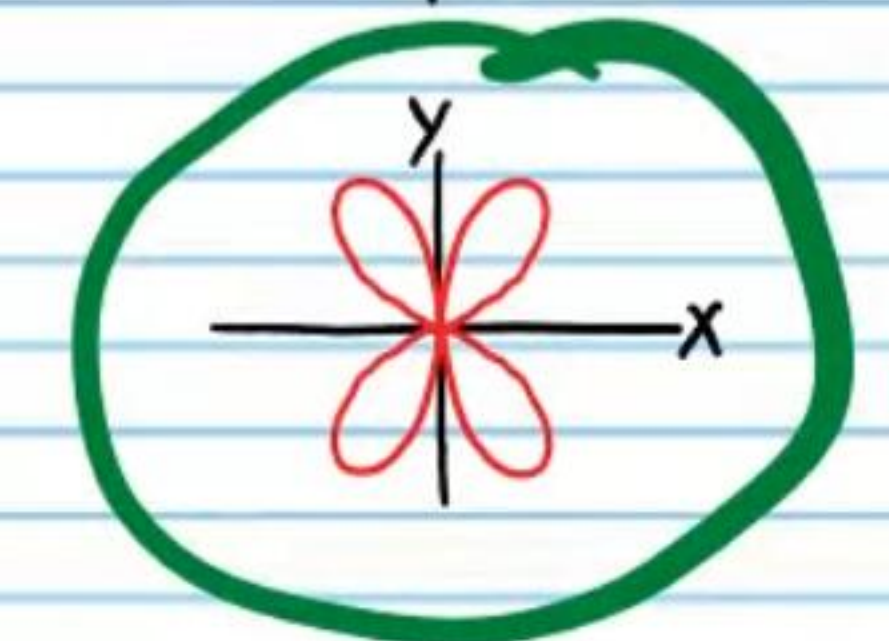
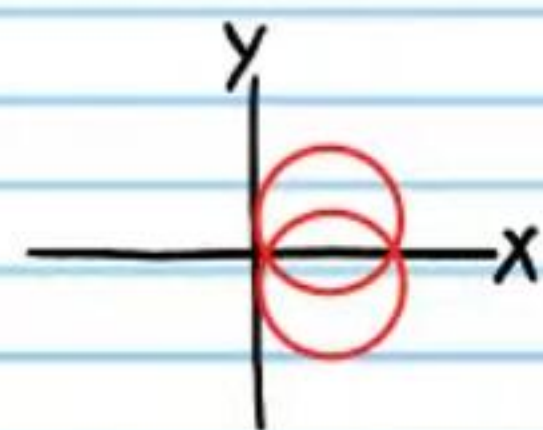
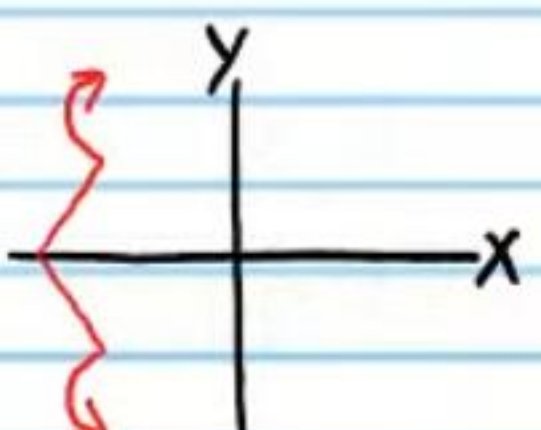
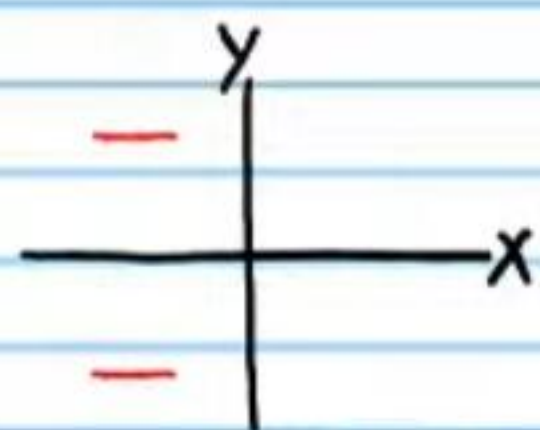
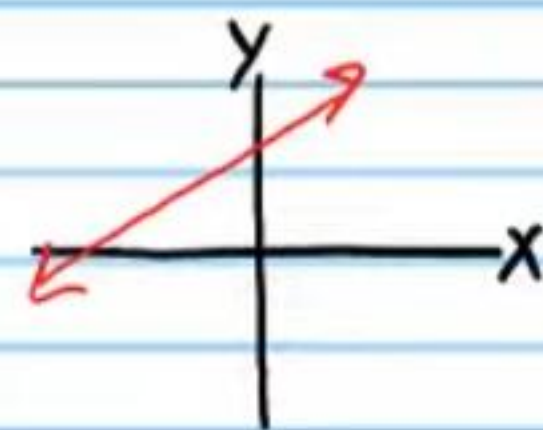
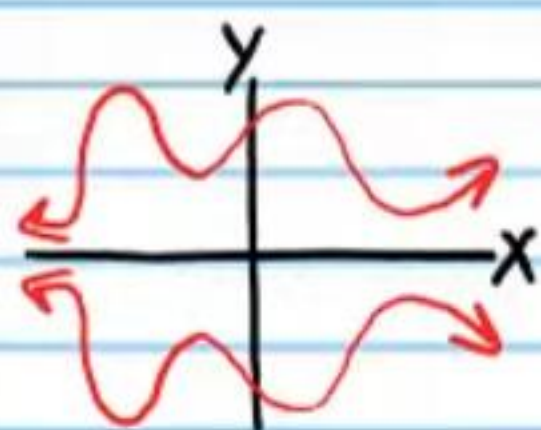
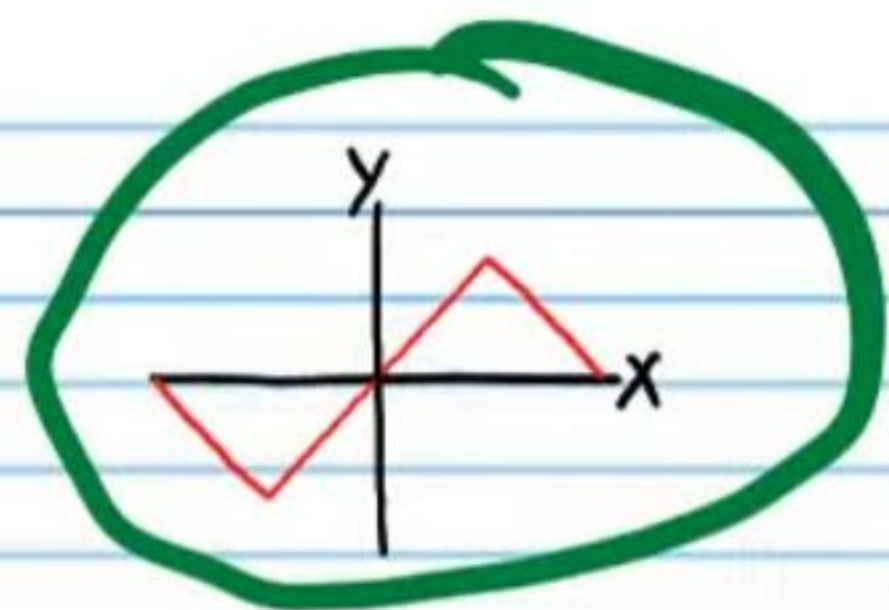
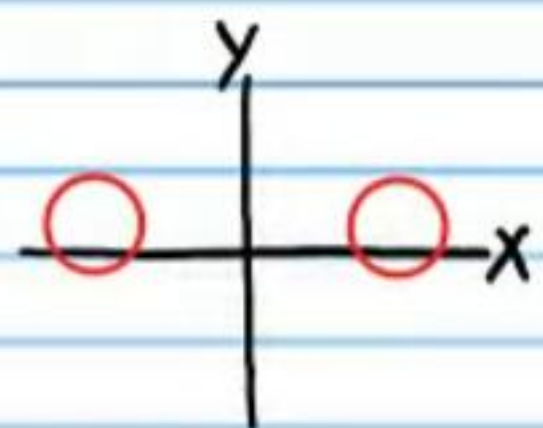
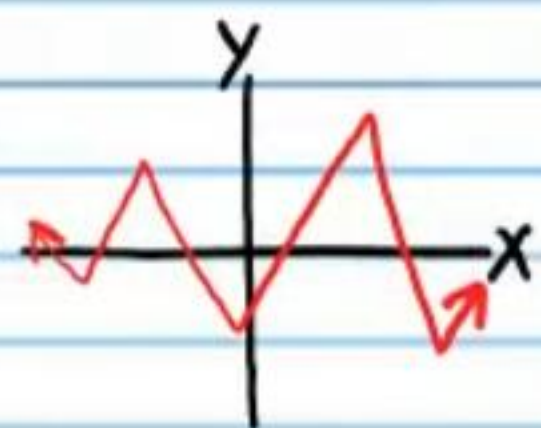
|   |    |     |     |    |     |     |     |      |     |    |
|---|----|-----|-----|----|-----|-----|-----|------|-----|----|
| X | 0  | 87  | -4  | -3 | 5   | 0   | -87 | 4    | 3   | -5 |
| Y | 15 | -20 | 100 | 33 | -14 | -15 | 20  | -100 | -33 | 14 |

43

|   |     |    |    |    |     |     |     |    |     |    |
|---|-----|----|----|----|-----|-----|-----|----|-----|----|
| X | 10  | 34 | 2  | 16 | -19 | -10 | -34 | -2 | -16 | 19 |
| Y | -10 | 30 | -2 | 10 | -10 | 10  | -30 | 2  | -10 | 10 |







④⑤  $\cos x - \sin x = y$

$\cos x + \sin x = -y$

$-1(\cos x + \sin x) = (-y)(-1)$

$-\cos x - \sin x = y$

✗

④⑥  $|x| = |y|$

$|-x| = |-y|$

$|x| = |y|$

✓

④⑦  $4\sqrt{-x-1} + 2 = -y$

✗

④⑧  $-\csc x + 5y = -x^7$

$-\csc(-x) + 5(-y) = -(-x)^7$

$-(-\csc x) - 5y = -(-x^7)$

$\csc x - 5y = x^7$

$-1(\csc x - 5y) = (x^7)(-1)$

$-\csc x + 5y = -x^7$

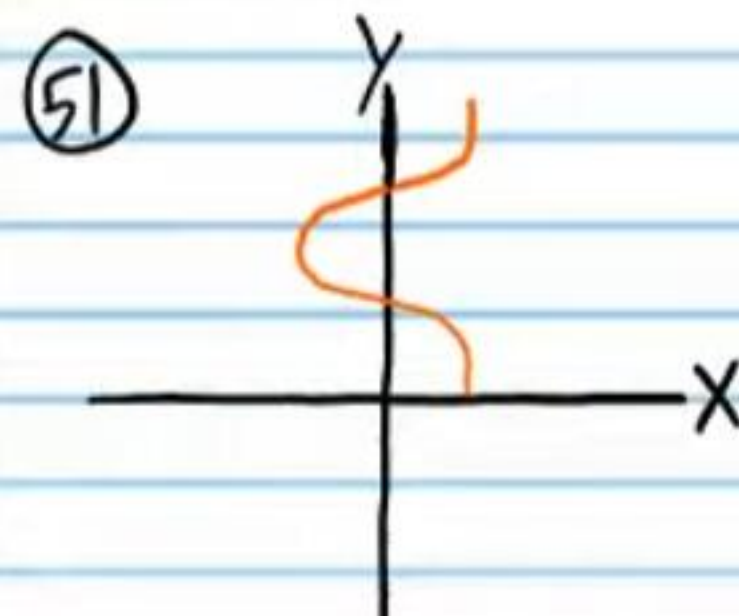
✓

④⑨

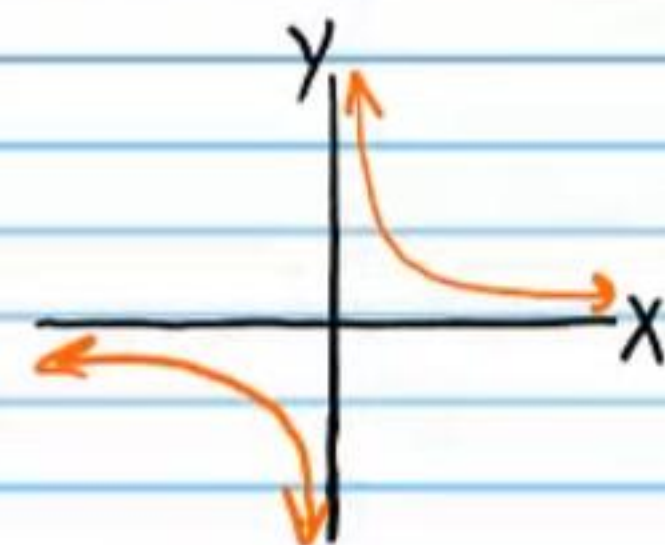
|   |    |    |     |    |    |
|---|----|----|-----|----|----|
| x | -7 | 3  | 2   | 0  | -8 |
| y | -5 | 16 | -16 | -6 | -9 |

⑤⑩

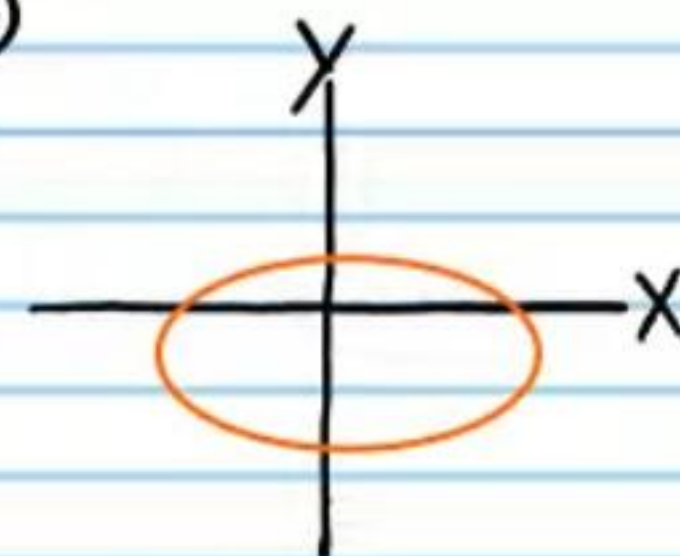
|   |    |    |    |    |    |
|---|----|----|----|----|----|
| x | 40 | 30 | 20 | 10 | 0  |
| y | 50 | 60 | 70 | 80 | 90 |



⑤②



⑤③



⑤④  $y + x = 1$

⑤⑤  $|y| = |x|$

⑤⑥  $y^3 + 10y^2 = x^2$

⑤⑦  $4^x - 2^{x-1} = y$